

```

1  /*****
2  * 0. Set library
3  *****/
4  libname eddy "/home/u64020665";
5
6  /*****
7  * 1. Construct long-format dataset + create binary outcome YBIN
8  *****/
9  data eddy.alz_long2;
10     set eddy.alz_wide;
11
12     array bprs_arr[7] bprs0-bprs6;
13     array cdrsb_arr[7] cdrsb0-cdrsb6;
14
15     do time = 0 to 6;
16
17         time2 = time*time; /* quadratic time term */
18
19         BPRS = bprs_arr[time+1];
20         CDRSB = cdrsb_arr[time+1];
21
22         /* Missingness indicators (Q1) */
23         miss_BPRS = (BPRS = .);
24         miss_CDRSB = (CDRSB = .);
25
26         /* -----
27         Binary outcome (Q3): define YBIN from CDRSB
28         CHANGE THIS to match your Assignment 2 definition.
29         Example below: YBIN = 1 if CDRSB >= 4, else 0
30         ----- */
31         if CDRSB = . then YBIN = .;
32         else YBIN = (CDRSB >= 4);
33
34         output;
35     end;
36
37     keep patid trial sex age edu bmi inkomen job adl wzc
38         abpet0 tauPET0
39         time time2 BPRS CDRSB YBIN miss_BPRS miss_CDRSB;
40 run;
41
42 /*****
43 * Empirical variance by time
44 *****/
45
46 proc means data=eddy.alz_long2 n mean var;
47     class time;
48     var BPRS;
49     output out=emp_var(drop=_type_ _freq_) n=n mean=mean var=emp_var;
50 run;
51
52 proc print data=emp_var noobs;
53 run;
54
55 /*****
56 * Fit final LMM
57 *****/
58 ods output CovParms=cov_bprs;
59

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60 proc mixed data=eddy.alz_long2 method=reml;
61   class patid sex;
62   model BPRS =
63     time time2
64     age sex bmi adl abpet0 tauPET0
65     time*age time*sex time*adl
66     / solution ddfm=kr;
67   random intercept time / subject=patid type=un;
68 run;
69
70 ods output close;
71
72 /*****
73 * compute the model-implied marginal variance by time
74 *****/
75
76 data _null_;
77   set cov_bprs;
78   /* Random effects UN(1,1), UN(2,1), UN(2,2) plus residual */
79   if CovParm="UN(1,1)" then call symputx('var_b0', Estimate);
80   if CovParm="UN(2,2)" then call symputx('var_b1', Estimate);
81   if CovParm="UN(2,1)" then call symputx('cov_b0b1', Estimate);
82
83   /* Residual variance: often shown as "Residual" */
84   if upcase(CovParm)="RESIDUAL" then call symputx('sigma2', Estimate);
85 run;
86
87 %put &=var_b0 &=var_b1 &=cov_b0b1 &=sigma2;
88
89 data fit_var;
90   set emp_var;
91   /* Model-implied marginal variance at each time */
92   fit_var = &var_b0 + 2*time*&cov_b0b1 + (time*time)*&var_b1 + &sigma2;
93 run;
94
95 proc print data=fit_var noobs;
96   var time emp_var fit_var;
97 run;
98
99 /*****
100 * plot empirical vs fitted variance curves
101 *****/
102
103 proc sgplot data=fit_var;
104   series x=time y=emp_var / legendlabel="Empirical variance";
105   series x=time y=fit_var / legendlabel="Model-implied variance";
106   xaxis integer label="Time";
107   yaxis label="Variance(BPRS)";
108 run;
109
110 /*****
111 * test heteroscedastic residual model
112 *****/
113
114 /* 1) Fit heteroscedastic residual LMM and capture CovParms */
115 ods output CovParms=cov_bprs_het;
116
117 proc mixed data=eddy.alz_long2 method=reml;
118   class patid sex time;
119   model BPRS =
120     time time2

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120     age sex bmi adl abpet0 taupe0
121     time*age time*sex time*adl
122     / solution ddfm=kr;
123
124     random intercept time / subject=patid type=un;
125
126     /* different residual variance per time */
127     repeated / subject=patid group=time type=vc;
128 run;
129
130 ods output close;
131
132 /* quick check that it exists */
133 proc contents data=cov_bprs_het; run;
134 proc print data=cov_bprs_het(obs=15); run;
135
136 data _null_;
137     set cov_bprs_het;
138     if CovParm="UN(1,1)" then call symputx('var_b0', Estimate);
139     if CovParm="UN(2,2)" then call symputx('var_b1', Estimate);
140     if CovParm="UN(2,1)" then call symputx('cov_b0b1', Estimate);
141 run;
142
143 %put &var_b0 &var_b1 &cov_b0b1;
144
145 data resid_var;
146     set cov_bprs_het;
147     where upcase(CovParm)="RESIDUAL";
148
149     /* Group looks like: "time 0", "time 1", ... */
150     time = input(scan(Group, 2, ' '), best.);
151
152     sigma2_t = Estimate;
153
154     keep time sigma2_t;
155 run;
156
157 proc sort data=resid_var; by time; run;
158
159
160 data fit_var_het;
161     merge emp_var resid_var;
162     by time;
163
164     fit_var =
165         &var_b0
166         + 2*time*&cov_b0b1
167         + (time*time)*&var_b1
168         + sigma2_t;
169 run;
170
171 proc sgplot data=fit_var_het;
172     series x=time y=emp_var / legendlabel="Empirical variance";
173     series x=time y=fit_var / legendlabel="Model-implied variance (heteroscedastic)";
174     xaxis integer label="Time";
175     yaxis label="Variance(BPRS)";
176 run;
177
178
179 proc print data=cov_bprs_het;
180     where upcase(CovParm)="RESIDUAL";

```

```
180   var CovParm Group Estimate;  
181 run;  
182  
183  
184  
185  
186  
187  
188  
189  
190  
191  
192  
---
```